SS-S-721c
November 18, 1964

SUPERSEDING
Int. Fed. Spec. SS-S-00721b
October 2, 1963 and
Fed. Spec. SS-S-721
November 10, 1931

FEDERAL SPECIFICATION

STONE, ARCHITECTURAL, CAST

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers material known as architectural cast stone, a building stone manufactured from portland cement concrete, precast and set in place as trim or facing on or in buildings, and other structures.

1.2 Classification.

1.2.1 *Types.* Cast stone shall be of the following types, as specified (see 6.2):

Type I—Cast stone, either homogeneous throughout, or with not less then 7/8 inch of facing.
Type II—Exposed aggregate masonry panels.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specification and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specification:

SS-C-192—Cement, Portland.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Certification. The contractor shall furnish certification that similar cast stone made from materials being used to meet the requirements of this specification have been used as exposed outdoor trim for ten

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years without signs of deterioration and surface defects which would affect its appearance.

3.2 Materials. The manufacturer is given a wide range in the selection of materials and processes of manufacture in order that cast stone of the prescribed quality will be produced. The cast stone shall be made with portland cement conforming to SS-C-192, or white, portland cement and aggregate having a specific gravity of not less than 2.60, unless proven to be durable by actual freezing and thawing test. Siliceous stone, marble, quartz, granite or any other natural stone used as trim or facing, exposed outdoors for ten or more years without signs of deterioration will be acceptable as fulfilling requirements when used as aggregate for cast stone.

3.3 Surface finish. All visible surfaces of cast stone, when it is in place, shall be free from surface defects and shall be finished to a minimum of 1/8 inch when manufactured using the aggregates and surface appearances listed in tables I and II and specified in 6.2. If a aggregate or surface finish other than those listed in tables I and II is required the aggregate and surface appearance shall be as specified (see 6.2).

TABLE I.—*Type I, aggregates and final surface appearances*

Aggregate	Final surface appearance
Siliceous stones, marble, granite, or quartz	Smooth uniform surface. Surface broken to give a very fine grained texture similar to natural stone.
	Ground surface, not polished.
Marble or granite	Polished surface.
	Tooled surface.

3.4 Reinforcement. Cast stone shall be reinforced with steel when necessary for safe handling and erection and the size and type of reinforcing shall be specified, (see 6.2). If concrete surfaces are to be exposed to weather or to be in contact with the ground, the reinforcement shall be protected with not less than 2 inches of concrete for bars larger than 5/8 inch and 1-1/2 inches for 5/8-inch bars or smaller. The concrete protective covering for reinforcement of surfaces that are not exposed directly to the ground or weather shall be not less than 3/4 inch for slabs and walls, and shall in all other cases be at least equal to the diameter of the bars. Exposed aggregate precast panels shall have a minimum thickness of 2 inches and reinforced with zinc-coated welded wire fabric and such additional reinforcement as may be required for handling, and to allow for temperature changes and structural stress, there shall be a minimum steel reinforcement amounting to 1/4 of 1 percent of the sectional area of the stone, should the stone be greater than 12 inches in any dimension the temperature steel shall be placed in both directions.

3.5 Sills and copings. Exterior sills, copings, projecting courses, and pieces with exposed top surfaces shall be formed with a wash. Where other work is built upon such pieces, raised seat or lugs shall be provided to form a bed for such work. Unless otherwise specified (see 6.2) window sills shall have raised fillets at the back. Pieces that project beyond the face of the work below shall have drips under the outer edge.

TABLE II.—*Type II, aggregate and final surface appearance*

Aggregate	Final surface appearance[1]
Siliceous stone, marble, or quartz	Fine aggregate 1/8 inch or less with fine texture.
	Aggregates 1/4 inch to 3/8 inch with exposures to cover 85 percent of panel face.
	1/2 inch to 3/4 inch aggregates exposed to cover 85 percent of panel face.
	3/4 inch to 1-1/2 inches aggregates exposed to cover 85 percent of the panel face.
	1-1/2 inches to 2-1/2 inches aggregates exposed to cover 85 percent of the panel face.

[1] In all cases not less than 85 percent of the aggregate faces shall be covered by the aggregates used.

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3.6 Lifting hook and anchorage holes. Holes for lifting and anchoring devices, such as linkages, anchors, or dowels, shall be cast into the stone wherever required, and shall be of the types and locations that will be concealed when stone is set. Setting loops or lifting hooks shall be placed at the time of casting. All holes and devices shall be at least one inch from finished wall.

3.7 Compressive strength. The average compressive strength of cast stone shall be 6,500 p.s.i. or greater, when cured for a minimum of 28 days, and subjected to the test in 4.4.2.

3.8 Absorption.

3.8.1 *Cold water.* The average absorption of water by the stone castings when immersed for 48 hours in water at 21° ± 3°C., shall be less than 5 percent by weight, when determined in accordance with 4.4.3.

3.8.2 *Boiling water.* The average absorption by the cast stone, after immersion in cold water as specified in 3.8.1 and subsequent boiling for 5 hours, shall be less than 10 percent, when tested in accordance with 4.4.3.

3.8.3 *Ratio of cold to boiling water absorption.* If the average total absorption after boiling as calculated in 4.4.3 for 3.8.2 is greater than 6 percent by weight the ratio of the total absorption as calculated in 4.4.3 for 3.8.1 to the absorption as calculated in 4.4.3 shall be less than 0.080.

3.9 Workmanship. Cast stone shall be free from defects such as cracks, loose aggregate, broken edges, and marred finish surfaces which may affect its appearance or which may impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Unless otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 *Lot.* Unless otherwise specified (see 6.2), a lot shall consist of all cast stone of the same type offered for inspection at the same time.

4.2.2 *Dimensional and visual examination of end item.* Sampling of cast stone for these characteristics shall be in accordance with MIL-STD-105, level S-3.

4.2.3 *Sampling for test.* For each 400 cubic feet of cast stone, or minimum of one set of test under 400 square feet submitted, samples for test shall be obtained from one unit or suitable size cubes cast from the same materials especially prepared for obtaining test specimens.

4.3 Inspection. Cast stone selected in 4.2.2 shall be examined in accordance with table III. The acceptable quality levels (AQL's) for defects listed in table III shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects in accordance with MIL-STD-105.

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TABLE III.—*Classification of defects*

Examine	Defects	Major	Minor
End item	Type not as specified.	X	
	Aggregate not as specified.	X	
	Finish not as specified.	X	
	Size not as specified.	X	
	Raised fillets missing on window sills.	X	
	Drips missing on projecting members.	X	
	Lifting or anchoring devices not flush.	X	
	Broken edges.	X	
	Damage or defective not affecting serviceability.		X
	Damage or defects affecting serviceability or appearance.	X	
Marking	Marking missing, incorrect, or illegible.		X

4.4 Tests.

4.4.1 *Specimens.*

4.4.1.1 *Compressive and absorption specimens.* Three specimens for absorption test and three specimens for compressive test shall be taken from each sample (see 4.2.3). They may be taken from any portion of the sample panel, except that for faced cast stone the specimens shall be cut through the faced surface, and the axis of the specimen perpendicular to the surface or specimen may be suitable size cubes compatible to the size aggregate used and shall consist of the facing mix and backup material, and the applied testing pressure shall be parallel to the finished surface. The AQL for test conducted with the specimens shall be 10.0 percent defective in accordance with MIL-STD-105.

4.4.1.2 *Bearing surfaces.* The bearing surfaces of specimens for strength test shall be rough ground and capped with leadite to produce an equivalent bearing surface to a deviation no greater than 0.002 inch. The leadite shall be as thin as practicable and in no case over 1/16 inch thick.

4.4.2 *Compressive.*

4.4.2.1 *Method.* The three compressive test specimens made as indicated in 4.4.1 shall be dried for 48 hours in an oven maintained at a temperature of 105°-110°C. They shall be removed from the oven, allowed to cool in laboratory air for 4 to 6 hours, and then tested in compression. A spherical bearing block shall be used between the upper face of the specimen and the upper platen; the center of the sphere of the block shall lie in the axis of the test cylinder and in the bearing face. The upper portion of the spherical bearing shall be slowly rotated as the head of the machine is brought to a bearing on the test cube. Cushioning materials shall not be used. Load shall be applied uniformly, without shock. The head of the testing machine shall move when idle at the speed of 0.05-inch-per-minute or the load shall be applied at the rate of 50-pounds-per-second.

4.4.2.2 *Calculation.* The specimens shall be loaded to failure and the compressive strength calculated in pounds per square inch. The compressive strength shall conform to the requirements of 3.7 when subjected to this test.

4.4.3 *Absorption.*

4.4.3.1 *Method.* The test specimens, after cutting, shall be dried in an oven maintained at 105°-110°C. until the loss in weight in 24 hours is less than 0.1 percent. The specimens shall be cooled in air for 4 to 6 hours, weighed, then immersed in water at 21° ± 3°C. for 48 hours, removed, the surface water wiped off with a damp cloth, then weighed with a balance which has an accuracy of within 0.1 gram. The specimen is then immersed in water at a temperature of 21° to 30°C., the water brought to a boil within 1 hour and kept boiling continually for 5 hours and the source of heat re-

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moved. The specimen shall be removed from the water when it has cooled to 30° or below, the surface water removed with a damp cloth, then again weighed. The material shall conform to the requirements of 3.8 when subjected to this test.

4.4.3.2 The 48 hour cold water absorption percent referred to in 3.8.1 is calculated as follows:

$$\text{Percent cold water absorption} = \frac{\text{Wt. after 48-hour absorption} - \text{Dry wt.} \times 100}{\text{Dry wt.}}$$

4.4.3.3 The total absorption percent referenced in 3.8.2 is calculated as follows:

$$\text{Percent total absorption} = \frac{\text{Wt. after boiling} - \text{Dry wt.} \times 100}{\text{Dry wt.}}$$

4.5 Rejection and retests. In case of failure of the test specimens to meet the requirements, samples may be resubmitted for test provided the producer has reworked the lots, as necessary to correct the deficiencies or has removed the nonconforming material.

4.6 Examination of preparation for delivery. The packing and marking of cast stone shall be examined to determine compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing. Cast stone shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking specified in the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to any special marking specified in the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Architectural cast stone covered by this specification is intended as a trim or facing on or in buildings and other structures.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type of cast stone required (1.2.1).
(*c*) Aggregate and method of finish required if different from 3.3.
(*d*) Aggregate and method of finish required (3.3).
(*e*) Type and size of reinforcement if required (3.4).
(*f*) Sill, if different from 3.5.
(*g*) Size of lot, if different from 4.2.1.

6.3 Transportation description. The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Stone, cast, not otherwise indexed by name, honed.
Carload minimum weight 36,000 pounds.

Motor:

Stone, cast, not otherwise indexed by name, honed.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY INTERESTS:

Army—MO

Navy—YD

Air Force—69

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